Program : **Diploma in Tool & Die Engineering**	
Course Code : **5111**	Course Title: **Tool Design**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To impart basic knowledge of kinetic link mechanism and degrees of freedom
- To familiarize students with design techniques of press tools and moulds
- To provide the basic understanding of jigs and fixtures used in tool making industry

Course Prerequisites:

Topic	Course code	Course name	Semester
Plastics		Applied Chemistry	1
Production processes		Production Engineering I &II	2 & 3
Mould basics		Moulds And Dies Technology	3
Press work operations		Press Tool Technology	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe factors governing the design of machine elements, basics of Kinetic link, design of threaded fasteners and basics of die Design	15	Understanding
CO2	Apply the concept of various die design and its components	18	Applying
CO3	Explain the concept of various mould design and its components	14	Understanding
CO4	Recognize various Jigs and Fixtures, locators and clamps	11	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2		3					
CO3		2					
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe factors governing the design of machine elements, basics of Kinetic link, design of threaded fasteners and basics of die Design**		
M1.01	Explain the basics of kinetic link and mechanism	4	Understanding
M1.02	Familiar with thread type fasteners	3	Understanding
M1.03	Explain the basics of die design and die components	4	Understanding
M1.04	Describe different die operations	4	Understanding
Contents: **General Considerations in Machine Design** Kinetic Link: Basic link motion, Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom Temporary fastener type: Thread: thread type, types of thread, forms of thread, Designation and nomenclature of screw thread **Introduction to die design** - die components - die operations - Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming Die classification - Simple Die, Compound Die, Progressive Die, Combination Die			
CO2	**Apply the concept of die design and its components**		
M2.01	Explain stepwise design procedure of a die	4	Understanding
M2.02	Apply design knowledge of compound tool	4	Applying
M2.03	Apply design knowledge of different types of bending tool	5	Applying
M2.04	Apply the design aspects of drawing tool	5	Applying
	Series Test I	1	

Contents:

Blanking force calculation

Steps to design a die - scrap strip - die block - blanking punch - piercing punch - punch plate - pilots - gages - finger stop - automatic stop - stripper - fasteners - die set - dimensions and notes-bill of materials (simple calculation)

Design of Compound tools - With Direct knock out - with indirect knockout.

Design of Bending Tools :- V-Bending Tool, U- Bending Tool, L-Bending Tool, curling tool, forming tool, Bending stress - calculation, Bending radius - calculation (simple numerical problems)

Design of Drawing tool, Deep draw tool, Inverted draw tool Factors to be considered while designing calculation of drawing force - calculation for cylindrical cup drawing press capacity, Blank holding force - calculation.(simple numerical problems)

CO3	**Explain the concept of mould design and its components**		
M3.01	Describe basics of mould and its terminology	4	Understanding
M3.02	Explain different types of plastic materials and its flow properties	2	Understanding
M3.03	Describe stepwise design of injection mould	4	Understanding
M3.04	Demonstrate basic design steps for compression mould, transfer mould, blow mould and die casting die	4	Understanding

Contents:

Definition of mould: Terminology - mould materials - plastic materials - types of plastics-melting - solidification - shrinkage - melt flow index - variable molecular weight - melt temperature - selection of injection speed - design check list - step by step design - split line - gating - ejection - cavity inserts - venting - water cooling - impression center-mould layout - main sectional view

Design of Compression moulds: For component showing Horizontal flash, For component showing vertical flash.

Design of Transfer mould: Pot transfer mould - plunger transfer mould.

Design of die-casting dies: For cold chamber system - for hot chamber system.

CO4	**Recognize suitable Jigs and Fixtures, locators and clamps**		
M4.01	Recognize appropriate Jigs and fixtures	3	Understanding
M4.02	Describe design aspect of locators	3	Understanding
M4.03	Select suitable clamps	2	Understanding
M4.04	Explain different types of fixtures	3	Understanding
	Series Test II	1	

Contents: **Jigs and Fixture**: Materials used in Jigs and fixtures Design of locators - principles of degrees of freedom and constraints. **Design of clamp**: principles and classification of clamps **Drilling Jigs**: Types - Components **Milling fixture**: Salient features and classification Other types of fixture - turning fixtures, grinding fixtures, welding fixtures and modular fixtures

Text/ Reference:

T/R	Book Title/Author
T1	Introduction to injection moulding - Pye
T2	Jigs and Fixtures - Joshi
T3	Press Tools: Design and Construction - Joshi.P.H.
R1	Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Hill
R2	Design of Jigs fixtures and press tools - v balachandran
R3	The Mould Design Guide - Peter Jones
R4	Plastic technology handbook - Donald E Hudgin
R5	Jigs and Fixtures - Hiran.E.Grant

Online Resources:

Sl.No	Website Link
1	https://myplasticmold.com/
2	https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-
3	https://www.misumi-techcentral.com
4	http://www.wisetool.com
5	https://www.thefabricator.com